AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 3b; due Tuesday, October 27, 2026

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[1] Let p be a monic polynomial of odd degree $2k+1$. This means that

$$p(x) = \sum_{j=0}^{2k+1} c_j x^j$$

with $c_j \in \mathbb{R}$ for $j \in \{0, 1, \ldots, 2k+1\}$ and $c_{2k+1} = 1$. Let $M = \max_{0 \leq j \leq 2k+1} |c_j| \geq 1$.

[a] Show that $p(2M) > 0$.

Hint: You will need to use the fact that $\sum_{j=0}^{2k} 2^j = \frac{1-2^{2k+1}}{1-2} = 2^{2k+1} - 1$.

[b] Show that $p(-2M) < 0$.

[c] Prove that there exists at least one point $x_o \in \mathbb{R}$ such that $p(x_o) = 0$.

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[2] Suppose that the surface of the Earth can be reasonably well-approximated to be a sphere $S = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = R\}$ for some $R > 0$. It is intuitively clear, and not hard to prove, that S is *path-connected*. You do not have to show this.

Let $T\colon S \to \mathbb{R}$ be the temperature function on the surface of the Earth. That is, $T(\mathbf{x})$ is the temperature at the point $\mathbf{x} \in S$. Assume that T is a *continuous* function. Show that there must exist at least one point $\mathbf{p}$ on the surface of the Earth which is at exactly the same temperature at the *antipodal* point $-\mathbf{p}$. That is, prove that there exists $\mathbf{p} \in S$ such that $T(\mathbf{p}) = T(-\mathbf{p})$.

Hint: Consider the function $f\colon S \to \mathbb{R}$ given by $f(\mathbf{x}) = T(\mathbf{x}) - T(-\mathbf{x})$.

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[3] Let $f\colon (0, \infty) \to \mathbb{R}$ be given by $f(x) = \frac{1}{x}$. We proved in class that f is continuous on $(0, \infty)$.

[a] Show that f is *not* uniformly continuous on $A = (0, 1)$.

[b] Show that f *is* uniformly continuous on $B = [1, \infty)$. [Note that B is *not compact*, so you cannot use the theorem from class. You need to do this directly.]

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[4] Fix a subset $E \subseteq \mathbb{R}^n$. Let $f, g\colon E \to \mathbb{R}^m$.

[a] Suppose that f and g are both uniformly continuous on E. Let $\lambda, \mu \in \mathbb{R}$. Show that $\lambda f + \mu g$ is also uniformly continuous on E.

[This shows that the space of functions from E to $\mathbb{R}^m$ which are uniformly continuous on E is a linear subspace of the vector space of all continuous functions from E to $\mathbb{R}^m$.]

[b] Let $m = 1$, so $fg\colon E \to \mathbb{R}$ is defined. Suppose that f and g are both uniformly continuous on E, and that they are both *bounded*. Show fg is uniformly continuous on E.

[This shows that the space of *bounded* uniformly continuous functions from E to $\mathbb{R}$ is closed under multiplication. In the language of abstract algebra, it is a *subring* of the ring of continuous functions from E to $\mathbb{R}$.]

[c] Show that the assumption that f, g are bounded is necessary for part [b]. That is, find examples of $f, g\colon E \subseteq \mathbb{R}^n \to \mathbb{R}$ which are uniformly continuous on E, and such that at least one of f, g is unbounded, and such that fg is not uniformly continuous on E. [Take $n = 1$ for simplicity.]

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[5] Suppose that $f\colon \mathbb{R}^n \to \mathbb{R}$ is continuous on $\mathbb{R}^n$ and that f *goes to zero at infinity* in the sense of problem [5] from Assignment #3a. Prove that f is uniformly continuous on $\mathbb{R}^n$.
